

# Received Order Analysis Engine (ROAE)

Analysis of the King Wen sequence including observations by Terence McKenna

## Hexagram reference table

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Hexagram reference table  
Each hexagram is a stack of 6 lines, either solid (1/yang) or broken (0/yin).  
The 'binary' column encodes these 6 lines as bits (bottom line = rightmost bit).  
Each hexagram is also composed of two 3-line trigrams: an upper and a lower.  
There are 8 possible trigrams (Heaven, Earth, Thunder, Water, Mountain, Wind, Fire, Lake), giving 8x8 = 64 possible hexagrams.  
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| Pos | Hex | Binary | Upper        | Lower        | Name                   |
|-----|-----|--------|--------------|--------------|------------------------|
| 01  | ䷀   | 111111 | Qian Heaven  | Qian Heaven  | Heaven over Heaven     |
| 02  | ䷁   | 000000 | Kun Earth    | Kun Earth    | Earth over Earth       |
| 03  | ䷂   | 010001 | Kan Water    | Zhen Thunder | Water over Thunder     |
| 04  | ䷃   | 100010 | Gen Mountain | Kan Water    | Mountain over Water    |
| 05  | ䷄   | 010111 | Kan Water    | Qian Heaven  | Water over Heaven      |
| 06  | ䷅   | 111010 | Qian Heaven  | Kan Water    | Heaven over Water      |
| 07  | ䷆   | 000010 | Kun Earth    | Kan Water    | Earth over Water       |
| 08  | ䷇   | 010000 | Kan Water    | Kun Earth    | Water over Earth       |
| 09  | ䷈   | 110111 | Xun Wind     | Qian Heaven  | Wind over Heaven       |
| 10  | ䷉   | 111011 | Qian Heaven  | Dui Lake     | Heaven over Lake       |
| 11  | ䷊   | 000111 | Kun Earth    | Qian Heaven  | Earth over Heaven      |
| 12  | ䷋   | 111000 | Qian Heaven  | Kun Earth    | Heaven over Earth      |
| 13  | ䷌   | 111101 | Qian Heaven  | Li Fire      | Heaven over Fire       |
| 14  | ䷍   | 101111 | Li Fire      | Qian Heaven  | Fire over Heaven       |
| 15  | ䷎   | 000100 | Kun Earth    | Gen Mountain | Earth over Mountain    |
| 16  | ䷏   | 001000 | Zhen Thunder | Kun Earth    | Thunder over Earth     |
| 17  | ䷐   | 011001 | Dui Lake     | Zhen Thunder | Lake over Thunder      |
| 18  | ䷑   | 100110 | Gen Mountain | Xun Wind     | Mountain over Wind     |
| 19  | ䷒   | 000011 | Kun Earth    | Dui Lake     | Earth over Lake        |
| 20  | ䷓   | 110000 | Xun Wind     | Kun Earth    | Wind over Earth        |
| 21  | ䷔   | 101001 | Li Fire      | Zhen Thunder | Fire over Thunder      |
| 22  | ䷕   | 100101 | Gen Mountain | Li Fire      | Mountain over Fire     |
| 23  | ䷖   | 100000 | Gen Mountain | Kun Earth    | Mountain over Earth    |
| 24  | ䷗   | 000001 | Kun Earth    | Zhen Thunder | Earth over Thunder     |
| 25  | ䷘   | 111001 | Qian Heaven  | Zhen Thunder | Heaven over Thunder    |
| 26  | ䷙   | 100111 | Gen Mountain | Qian Heaven  | Mountain over Heaven   |
| 27  | ䷚   | 100001 | Gen Mountain | Zhen Thunder | Mountain over Thunder  |
| 28  | ䷛   | 011110 | Dui Lake     | Xun Wind     | Lake over Wind         |
| 29  | ䷜   | 010010 | Kan Water    | Kan Water    | Water over Water       |
| 30  | ䷝   | 101101 | Li Fire      | Li Fire      | Fire over Fire         |
| 31  | ䷞   | 011100 | Dui Lake     | Gen Mountain | Lake over Mountain     |
| 32  | ䷟   | 001110 | Zhen Thunder | Xun Wind     | Thunder over Wind      |
| 33  | ䷠   | 111100 | Qian Heaven  | Gen Mountain | Heaven over Mountain   |
| 34  | ䷡   | 001111 | Zhen Thunder | Qian Heaven  | Thunder over Heaven    |
| 35  | ䷢   | 101000 | Li Fire      | Kun Earth    | Fire over Earth        |
| 36  | ䷣   | 000101 | Kun Earth    | Li Fire      | Earth over Fire        |
| 37  | ䷤   | 110101 | Xun Wind     | Li Fire      | Wind over Fire         |
| 38  | ䷥   | 101011 | Li Fire      | Dui Lake     | Fire over Lake         |
| 39  | ䷦   | 010100 | Kan Water    | Gen Mountain | Water over Mountain    |
| 40  | ䷧   | 001010 | Zhen Thunder | Kan Water    | Thunder over Water     |
| 41  | ䷨   | 100011 | Gen Mountain | Dui Lake     | Mountain over Lake     |
| 42  | ䷩   | 110001 | Xun Wind     | Zhen Thunder | Wind over Thunder      |
| 43  | ䷪   | 011111 | Dui Lake     | Qian Heaven  | Lake over Heaven       |
| 44  | ䷫   | 111110 | Qian Heaven  | Xun Wind     | Heaven over Wind       |
| 45  | ䷬   | 011000 | Dui Lake     | Kun Earth    | Lake over Earth        |
| 46  | ䷭   | 000110 | Kun Earth    | Xun Wind     | Earth over Wind        |
| 47  | ䷮   | 011010 | Dui Lake     | Kan Water    | Lake over Water        |
| 48  | ䷯   | 010110 | Kan Water    | Xun Wind     | Water over Wind        |
| 49  | ䷰   | 011101 | Dui Lake     | Li Fire      | Lake over Fire         |
| 50  | ䷱   | 101110 | Li Fire      | Xun Wind     | Fire over Wind         |
| 51  | ䷲   | 001001 | Zhen Thunder | Zhen Thunder | Thunder over Thunder   |
| 52  | ䷳   | 100100 | Gen Mountain | Gen Mountain | Mountain over Mountain |

|    |  |   |  |        |  |              |  |              |  |                       |
|----|--|---|--|--------|--|--------------|--|--------------|--|-----------------------|
| 53 |  | ䷶ |  | 110100 |  | Xun Wind     |  | Gen Mountain |  | Wind over Mountain    |
| 54 |  | ䷲ |  | 001011 |  | Zhen Thunder |  | Dui Lake     |  | Thunder over Lake     |
| 55 |  | ䷲ |  | 001101 |  | Zhen Thunder |  | Li Fire      |  | Thunder over Fire     |
| 56 |  | ䷶ |  | 101100 |  | Li Fire      |  | Gen Mountain |  | Fire over Mountain    |
| 57 |  | ䷶ |  | 110110 |  | Xun Wind     |  | Xun Wind     |  | Wind over Wind        |
| 58 |  | ䷲ |  | 011011 |  | Dui Lake     |  | Dui Lake     |  | Lake over Lake        |
| 59 |  | ䷶ |  | 110010 |  | Xun Wind     |  | Kan Water    |  | Wind over Water       |
| 60 |  | ䷲ |  | 010011 |  | Kan Water    |  | Dui Lake     |  | Water over Lake       |
| 61 |  | ䷶ |  | 110011 |  | Xun Wind     |  | Dui Lake     |  | Wind over Lake        |
| 62 |  | ䷲ |  | 001100 |  | Zhen Thunder |  | Gen Mountain |  | Thunder over Mountain |
| 63 |  | ䷲ |  | 010101 |  | Kan Water    |  | Li Fire      |  | Water over Fire       |
| 64 |  | ䷶ |  | 101010 |  | Li Fire      |  | Kan Water    |  | Fire over Water       |

## Reverse vs. Inverse pair analysis

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### Reverse vs. Inverse pair analysis

The 64 hexagrams are traditionally grouped into 32 pairs (1-2, 3-4, ..., 63-64).

For each pair, we test two relationships:

Reverse: flip the hexagram upside down (read the lines top-to-bottom instead of bottom-to-top). If the second hexagram upside-down equals the first, they are a reverse pair.

Inverse: toggle every line (solid becomes broken, broken becomes solid).

If toggling all lines of one hexagram gives the other, they are inverse.

Key finding: EVERY pair in the King Wen sequence is one or the other – none are unrelated. This perfect pairing structure is unlikely to occur by chance.

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01 ䷶ Inverse 02 ䷶  
03 ䷲ Reverse 04 ䷲  
05 ䷲ Reverse 06 ䷲  
07 ䷲ Reverse 08 ䷲  
09 ䷶ Reverse 10 ䷶  
11 ䷶ Reverse 12 ䷶  
13 ䷶ Reverse 14 ䷶  
15 ䷲ Reverse 16 ䷲  
17 ䷶ Reverse 18 ䷶  
19 ䷲ Reverse 20 ䷲  
21 ䷲ Reverse 22 ䷲  
23 ䷲ Reverse 24 ䷲  
25 ䷶ Reverse 26 ䷶  
27 ䷲ Inverse 28 ䷶  
29 ䷲ Inverse 30 ䷶  
31 ䷶ Reverse 32 ䷶  
33 ䷶ Reverse 34 ䷶  
35 ䷲ Reverse 36 ䷲  
37 ䷶ Reverse 38 ䷶  
39 ䷲ Reverse 40 ䷲  
41 ䷲ Reverse 42 ䷲  
43 ䷶ Reverse 44 ䷶  
45 ䷲ Reverse 46 ䷲  
47 ䷲ Reverse 48 ䷲  
49 ䷶ Reverse 50 ䷶  
51 ䷲ Reverse 52 ䷲  
53 ䷲ Reverse 54 ䷲  
55 ䷲ Reverse 56 ䷲  
57 ䷲ Reverse 58 ䷲  
59 ䷲ Reverse 60 ䷲  
61 ䷲ Inverse 62 ䷲  
63 ䷲ Reverse 64 ䷲

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Reverse count pairs: 28/32 (87.5%)  
Inverse count pairs: 04/32 (12.5%)  
Neither count pairs: 00/32 (0.0%)

## First order of difference (the 'wave')

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First order of difference (the 'wave')

For each consecutive pair of hexagrams, count how many of the 6 lines change.  
This produces a sequence of values from 0 (identical) to 6 (every line flipped).  
The resulting 'wave' is central to analysis of the King Wen sequence.  
Key observation (McKenna, The Invisible Landscape, 1975): the value 5 never appears – no consecutive hexagrams differ by exactly 5 lines. The value 0 also never appears (no duplicates).

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01 ䷀ to 02 ䷁ difference 6 *****
02 ䷁ to 03 ䷂ difference 2 **
03 ䷂ to 04 ䷃ difference 4 ****
04 ䷃ to 05 ䷄ difference 4 ****
05 ䷄ to 06 ䷅ difference 4 ****
06 ䷅ to 07 ䷆ difference 3 ***
07 ䷆ to 08 ䷇ difference 2 **
08 ䷇ to 09 ䷈ difference 4 ****
09 ䷈ to 10 ䷉ difference 2 **
10 ䷉ to 11 ䷊ difference 4 ****
11 ䷊ to 12 ䷋ difference 6 *****
12 ䷋ to 13 ䷌ difference 2 **
13 ䷌ to 14 ䷍ difference 2 **
14 ䷍ to 15 ䷎ difference 4 ****
15 ䷎ to 16 ䷏ difference 2 **
16 ䷏ to 17 ䷐ difference 2 **
17 ䷐ to 18 ䷑ difference 6 *****
18 ䷑ to 19 ䷒ difference 3 ***
19 ䷒ to 20 ䷓ difference 4 ****
20 ䷓ to 21 ䷔ difference 3 ***
21 ䷔ to 22 ䷕ difference 2 **
22 ䷕ to 23 ䷖ difference 2 **
23 ䷖ to 24 ䷗ difference 2 **
24 ䷗ to 25 ䷘ difference 3 ***
25 ䷘ to 26 ䷙ difference 4 ****
26 ䷙ to 27 ䷚ difference 2 **
27 ䷚ to 28 ䷛ difference 6 *****
28 ䷛ to 29 ䷜ difference 2 **
29 ䷜ to 30 ䷝ difference 6 *****
30 ䷝ to 31 ䷞ difference 3 ***
31 ䷞ to 32 ䷟ difference 2 **
32 ䷟ to 33 ䷠ difference 3 ***
33 ䷠ to 34 ䷡ difference 4 ****
34 ䷡ to 35 ䷢ difference 4 ****
35 ䷢ to 36 ䷣ difference 4 ****
36 ䷣ to 37 ䷤ difference 2 **
37 ䷤ to 38 ䷥ difference 4 ****
38 ䷥ to 39 ䷦ difference 6 *****
39 ䷦ to 40 ䷧ difference 4 ****
40 ䷧ to 41 ䷨ difference 3 ***
41 ䷨ to 42 ䷩ difference 2 **
42 ䷩ to 43 ䷪ difference 4 ****
43 ䷪ to 44 ䷫ difference 2 **
44 ䷫ to 45 ䷬ difference 3 ***
45 ䷬ to 46 ䷭ difference 4 ****
46 ䷭ to 47 ䷮ difference 3 ***
47 ䷮ to 48 ䷯ difference 2 **
48 ䷯ to 49 ䷰ difference 3 ***
49 ䷰ to 50 ䷱ difference 4 ****
50 ䷱ to 51 ䷲ difference 4 ****
51 ䷲ to 52 ䷳ difference 4 ****
52 ䷳ to 53 ䷴ difference 1 *
53 ䷴ to 54 ䷵ difference 6 *****
54 ䷵ to 55 ䷶ difference 2 **
55 ䷶ to 56 ䷷ difference 2 **
56 ䷷ to 57 ䷸ difference 3 ***
57 ䷸ to 58 ䷹ difference 4 ****
58 ䷹ to 59 ䷺ difference 3 ***
59 ䷺ to 60 ䷻ difference 2 **
60 ䷻ to 61 ䷼ difference 1 *
61 ䷼ to 62 ䷽ difference 6 *****
62 ䷽ to 63 ䷾ difference 3 ***
63 ䷾ to 64 ䷿ difference 6 *****
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```

0 line changes total 0  
1 line changes total 2  
2 line changes total 20  
3 line changes total 13  
4 line changes total 19

6 line changes total 9

| Age Group | Percentage |
|-----------|------------|
| 18-24     | 10         |
| 25-34     | 25         |
| 35-44     | 28         |
| 45-54     | 15         |
| 55-64     | 10         |
| 65-74     | 8          |
| 75-84     | 5          |
| 85-94     | 3          |
| 95+       | 2          |

Note: does not survive Bonferroni correction for 28 tests (threshold  $p < 0.0018$ ).

|      | Kun | Zhen | Kan | Dui | Gen | Li | Xun | Qian |
|------|-----|------|-----|-----|-----|----|-----|------|
| Kun  | 0   | 1    | 2   | 1   | 0   | 0  | 2   | 2    |
| Zhen | 0   | 1    | 1   | 1   | 2   | 2  | 0   | 1    |
| Kan  | 0   | 1    | 0   | 1   | 1   | 2  | 2   | 1    |
| Dui  | 1   | 1    | 2   | 0   | 1   | 1  | 1   | 1    |
| Gen  | 2   | 0    | 1   | 1   | 2   | 0  | 2   | 0    |
| Li   | 2   | 1    | 1   | 1   | 1   | 0  | 1   | 0    |
| Xun  | 0   | 2    | 1   | 2   | 0   | 2  | 0   | 1    |
| Qian | 3   | 1    | 0   | 1   | 1   | 1  | 0   | 1    |

```
--- Lower trigram transition matrix ---
Rows = from, Columns = to
      Kun   Zhen Kan   Dui   Gen   Li   Xun   Qian
Kun    0     4    0     0     0    2    1    1
Zhen   0     1    1     0     1    1    2    2
Kan    1     0    1     2     0    1    1    1
Dui    1     1    1     1     2    1    0    1
Gen    1     0    1     1     1    1    2    1
Li     1     0    1     1     2    1    1    1
Xun    1     1    2     2     1    1    0    0
Qian   3     1    1     1     1    0    1    0

--- Trigram change rates vs null (pair-preserving permutation test) ---
Null: 10,000 orderings preserving the pair structure (permute pairs + flip orientations),
the correct baseline per CRITIQUE.md. Statistic: upper/lower trigram change counts.
KW upper changes 59/63: percentile 47.3 (fraction of null <= KW)
KW lower changes 58/63: percentile 27.3

--- Pure (doubled-trigram) hexagram placement ---
The 8 pure hexagrams (upper == lower trigram). Lai Zhide (1525-1604, via Schulz 1982)
observed kan/li doubles closing both Classics; measured here against the same null.
Pure hexagram positions (1-based): [1, 2, 29, 30, 51, 52, 57, 58]
Pure hexagrams at Classic ends (positions 1,2,29,30,63,64): 4 of a possible 6
Null P(>= KW's 4) = 0.0336 (pair-preserving null; note KW's C4 fixes 1,2 by
definition, so interpret against the constrained baseline)

--- Nuclear trigram structure ---
Nuclear hexagram = lines 2-4 (lower nuclear) + 3-5 (upper nuclear). Classical fact
(commentary tradition): iterating the nuclear map sends all 64 hexagrams to 16, then to
the 4 fixed points 63, 0, and the two alternators 21(010101b rendering varies), 42.
Distinct after 1 application: 16; after 2: 4; after 3: 4 -> [0, 21, 42, 63]
KW nuclear-hexagram changes along the sequence: 58/63

--- Jing Fang Eight-Palaces rank correlation ---
Spearman rank correlation between KW positions and Jing Fang's palace ordering
(the trigram-generated classical alternative; ordering as in solve.c --null-historical).
Spearman rho = 0.1384; null P(|rho| >= observed) = 0.1170 (pair-preserving null)

--- Symmetry group vs the trigram decomposition (ROAE SYMMETRY_SEARCH theorem) ---
The C1-C5 symmetry group B3 (order 48) acts on LINE positions; reversal maps the upper
trigram to the reversed lower trigram, so most of the group does NOT preserve the
upper/lower split. Preserving elements = permutations acting within {0,1,2} and {3,4,5}
blocks (or swapping them wholesale) that also centralize rev.
Split-respecting subgroup order: 12 of 48 (computed over the rev-centralizer)

--- Methodological note ---
With ~1 expected observation per cell, no goodness-of-fit test (e.g.,
chi-square) has sufficient power to detect deviations from uniform
transitions. The matrices are descriptive only.
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# Nuclear hexagram analysis

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Nuclear hexagram analysis
Every hexagram contains a hidden 'nuclear' hexagram inside it, formed by
extracting the 4 middle lines (lines 2-3-4-5). Lines 2-3-4 form the lower
nuclear trigram, and lines 3-4-5 form the upper nuclear trigram (sharing lines
3 and 4). This creates a chain: each hexagram points to its nuclear hexagram,
which points to its own nuclear hexagram, and so on until the chain loops.
These chains reveal deep structural relationships between hexagrams that are
not obvious from the surface arrangement.
---
01 ䷁ nuclear: 01 ䷁ chain: 01 ䷁ -> 01 ䷁
02 ䷇ nuclear: 02 ䷇ chain: 02 ䷇ -> 02 ䷇
03 ䷊ nuclear: 23 ䷊ chain: 03 ䷊ -> 23 ䷊ -> 02 ䷇ -> 02 ䷇
04 ䷋ nuclear: 24 ䷋ chain: 04 ䷋ -> 24 ䷋ -> 02 ䷇ -> 02 ䷇
05 ䷌ nuclear: 38 ䷌ chain: 05 ䷌ -> 38 ䷌ -> 63 ䷌ -> 64 ䷌ -> 63 ䷌
06 ䷍ nuclear: 37 ䷍ chain: 06 ䷍ -> 37 ䷍ -> 64 ䷍ -> 63 ䷌ -> 64 ䷌
07 ䷎ nuclear: 24 ䷎ chain: 07 ䷎ -> 24 ䷎ -> 02 ䷇ -> 02 ䷇
08 ䷏ nuclear: 23 ䷏ chain: 08 ䷏ -> 23 ䷏ -> 02 ䷇ -> 02 ䷇
09 ䷐ nuclear: 38 ䷐ chain: 09 ䷐ -> 38 ䷐ -> 63 ䷌ -> 64 ䷌ -> 63 ䷌
```

|    |               |             |         |         |         |         |
|----|---------------|-------------|---------|---------|---------|---------|
| 10 | ☰ nuclear: 37 | ☷ chain: 10 | ☰ -> 37 | ☰ -> 64 | ☷ -> 63 | ☰ -> 64 |
| 11 | ☰ nuclear: 54 | ☷ chain: 11 | ☰ -> 54 | ☰ -> 63 | ☰ -> 64 | ☷ -> 63 |
| 12 | ☰ nuclear: 53 | ☷ chain: 12 | ☰ -> 53 | ☰ -> 64 | ☷ -> 63 | ☰ -> 64 |
| 13 | ☰ nuclear: 44 | ☷ chain: 13 | ☰ -> 44 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 14 | ☰ nuclear: 43 | ☷ chain: 14 | ☰ -> 43 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 15 | ☰ nuclear: 40 | ☷ chain: 15 | ☰ -> 40 | ☰ -> 63 | ☰ -> 64 | ☷ -> 63 |
| 16 | ☰ nuclear: 39 | ☷ chain: 16 | ☰ -> 39 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 17 | ☰ nuclear: 53 | ☷ chain: 17 | ☰ -> 53 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 18 | ☰ nuclear: 54 | ☷ chain: 18 | ☰ -> 54 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 19 | ☰ nuclear: 24 | ☷ chain: 19 | ☰ -> 24 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 20 | ☰ nuclear: 23 | ☷ chain: 20 | ☰ -> 23 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 21 | ☰ nuclear: 39 | ☷ chain: 21 | ☰ -> 39 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 22 | ☰ nuclear: 40 | ☷ chain: 22 | ☰ -> 40 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 23 | ☰ nuclear: 02 | ☷ chain: 23 | ☰ -> 02 | ☰ -> 02 | ☷       |         |
| 24 | ☰ nuclear: 02 | ☷ chain: 24 | ☰ -> 02 | ☰ -> 02 | ☷       |         |
| 25 | ☰ nuclear: 53 | ☷ chain: 25 | ☰ -> 53 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 26 | ☰ nuclear: 54 | ☷ chain: 26 | ☰ -> 54 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 27 | ☰ nuclear: 02 | ☷ chain: 27 | ☰ -> 02 | ☰ -> 02 | ☷       |         |
| 28 | ☰ nuclear: 01 | ☷ chain: 28 | ☰ -> 01 | ☰ -> 01 | ☷       |         |
| 29 | ☰ nuclear: 27 | ☷ chain: 29 | ☰ -> 27 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 30 | ☰ nuclear: 28 | ☷ chain: 30 | ☰ -> 28 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 31 | ☰ nuclear: 44 | ☷ chain: 31 | ☰ -> 44 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 32 | ☰ nuclear: 43 | ☷ chain: 32 | ☰ -> 43 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 33 | ☰ nuclear: 44 | ☷ chain: 33 | ☰ -> 44 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 34 | ☰ nuclear: 43 | ☷ chain: 34 | ☰ -> 43 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 35 | ☰ nuclear: 39 | ☷ chain: 35 | ☰ -> 39 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 36 | ☰ nuclear: 40 | ☷ chain: 36 | ☰ -> 40 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 37 | ☰ nuclear: 64 | ☷ chain: 37 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 | ☷       |
| 38 | ☰ nuclear: 63 | ☷ chain: 38 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 | ☷       |
| 39 | ☰ nuclear: 64 | ☷ chain: 39 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 | ☷       |
| 40 | ☰ nuclear: 63 | ☷ chain: 40 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 | ☷       |
| 41 | ☰ nuclear: 24 | ☷ chain: 41 | ☰ -> 24 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 42 | ☰ nuclear: 23 | ☷ chain: 42 | ☰ -> 23 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 43 | ☰ nuclear: 01 | ☷ chain: 43 | ☰ -> 01 | ☰ -> 01 | ☷       |         |
| 44 | ☰ nuclear: 01 | ☷ chain: 44 | ☰ -> 01 | ☰ -> 01 | ☷       |         |
| 45 | ☰ nuclear: 53 | ☷ chain: 45 | ☰ -> 53 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 46 | ☰ nuclear: 54 | ☷ chain: 46 | ☰ -> 54 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 47 | ☰ nuclear: 37 | ☷ chain: 47 | ☰ -> 37 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 48 | ☰ nuclear: 38 | ☷ chain: 48 | ☰ -> 38 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 49 | ☰ nuclear: 44 | ☷ chain: 49 | ☰ -> 44 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 50 | ☰ nuclear: 43 | ☷ chain: 50 | ☰ -> 43 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 51 | ☰ nuclear: 39 | ☷ chain: 51 | ☰ -> 39 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 52 | ☰ nuclear: 40 | ☷ chain: 52 | ☰ -> 40 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 53 | ☰ nuclear: 64 | ☷ chain: 53 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 | ☷       |
| 54 | ☰ nuclear: 63 | ☷ chain: 54 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 | ☷       |
| 55 | ☰ nuclear: 28 | ☷ chain: 55 | ☰ -> 28 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 56 | ☰ nuclear: 28 | ☷ chain: 56 | ☰ -> 28 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 57 | ☰ nuclear: 38 | ☷ chain: 57 | ☰ -> 38 | ☰ -> 63 | ☰ -> 64 | ☰ -> 63 |
| 58 | ☰ nuclear: 37 | ☷ chain: 58 | ☰ -> 37 | ☰ -> 64 | ☰ -> 63 | ☰ -> 64 |
| 59 | ☰ nuclear: 27 | ☷ chain: 59 | ☰ -> 27 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 60 | ☰ nuclear: 27 | ☷ chain: 60 | ☰ -> 27 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 61 | ☰ nuclear: 27 | ☷ chain: 61 | ☰ -> 27 | ☰ -> 02 | ☰ -> 02 | ☷       |
| 62 | ☰ nuclear: 28 | ☷ chain: 62 | ☰ -> 28 | ☰ -> 01 | ☰ -> 01 | ☷       |
| 63 | ☰ nuclear: 64 | ☷ chain: 63 | ☰ -> 64 | ☰ -> 63 | ☷       |         |
| 64 | ☰ nuclear: 63 | ☷ chain: 64 | ☰ -> 63 | ☰ -> 64 | ☷       |         |

--- Nuclear hexagram frequency ---

How often each hexagram appears as another's nuclear hexagram

|    |                         |                 |
|----|-------------------------|-----------------|
| 01 | ☰ Heaven over Heaven    | appears 4 times |
| 02 | ☷ Earth over Earth      | appears 4 times |
| 23 | ☶ Mountain over Earth   | appears 4 times |
| 24 | ☷ Earth over Thunder    | appears 4 times |
| 27 | ☶ Mountain over Thunder | appears 4 times |
| 28 | ☱ Lake over Wind        | appears 4 times |
| 37 | ☱ Wind over Fire        | appears 4 times |
| 38 | ☱ Fire over Lake        | appears 4 times |
| 39 | ☵ Water over Mountain   | appears 4 times |
| 40 | ☳ Thunder over Water    | appears 4 times |
| 43 | ☱ Lake over Heaven      | appears 4 times |
| 44 | ☰ Heaven over Wind      | appears 4 times |
| 53 | ☱ Wind over Mountain    | appears 4 times |
| 54 | ☳ Thunder over Lake     | appears 4 times |
| 63 | ☵ Water over Fire       | appears 4 times |
| 64 | ☱ Fire over Water       | appears 4 times |

--- Null model note ---  
Nuclear hexagram derivation is a fixed function of binary value (lines 2-5), independent of the King Wen ordering. The chain structure, cycle lengths, and frequency distribution above are identical for ANY ordering of the 64 hexagrams. These properties reflect the 6-bit binary system, not the King Wen sequence specifically.

## Line change positional analysis

---  
Line change positional analysis  
Each hexagram has 6 lines (positions 1-6, bottom to top). When consecutive hexagrams differ, which specific lines are changing? If the King Wen sequence were random, each line position would change about 50% of the time. Significant deviation from 50% would suggest the ordering favors changing certain positions. The co-change matrix shows which pairs of lines tend to flip together in the same transition – revealing correlated line movements.  
---  
Line 1: 37/63 ( 58.7%) #####  
Line 2: 32/63 ( 50.8%) #####  
Line 3: 35/63 ( 55.6%) #####  
Line 4: 34/63 ( 54.0%) #####  
Line 5: 35/63 ( 55.6%) #####  
Line 6: 38/63 ( 60.3%) #####

--- Line co-change frequency ---  
How often pairs of lines change together in the same transition

|    | L1 | L2 | L3 | L4 | L5 | L6 |
|----|----|----|----|----|----|----|
| L1 | .  | 18 | 20 | 19 | 19 | 30 |
| L2 | .  | .  | 20 | 17 | 23 | 20 |
| L3 | .  | .  | .  | 25 | 16 | 21 |
| L4 | .  | .  | .  | .  | 20 | 20 |
| L5 | .  | .  | .  | .  | .  | 20 |
| L6 | .  | .  | .  | .  | .  | .  |

## Complement distance analysis

---  
Complement distance analysis  
Every hexagram has an 'opposite' (complement) formed by toggling all 6 lines (solid becomes broken, broken becomes solid). For example, hexagram 1 (all solid) and hexagram 2 (all broken) are complements. This section finds where each hexagram's complement sits in the King Wen sequence and how far apart they are. If complements sit nearer each other than a null model predicts, opposition is an organizing feature of the sequence's structure; if farther, complements may serve as bookends for larger structural sections.  
---  
01 ☰ <-> 02 ☷ distance: 1 Heaven over Heaven <-> Earth over Earth  
02 ☷ <-> 01 ☰ distance: 1 Earth over Earth <-> Heaven over Heaven  
03 ☵ <-> 50 ☲ distance: 47 Water over Thunder <-> Fire over Wind  
04 ☴ <-> 49 ☶ distance: 45 Mountain over Water <-> Lake over Fire  
05 ☳ <-> 35 ☱ distance: 30 Water over Heaven <-> Fire over Earth  
06 ☱ <-> 36 ☵ distance: 30 Heaven over Water <-> Earth over Fire  
07 ☵ <-> 13 ☴ distance: 6 Earth over Water <-> Heaven over Fire  
08 ☴ <-> 14 ☳ distance: 6 Water over Earth <-> Fire over Heaven  
09 ☲ <-> 16 ☱ distance: 7 Wind over Heaven <-> Thunder over Earth  
10 ☶ <-> 15 ☴ distance: 5 Heaven over Lake <-> Earth over Mountain  
11 ☴ <-> 12 ☱ distance: 1 Earth over Heaven <-> Heaven over Earth  
12 ☱ <-> 11 ☴ distance: 1 Heaven over Earth <-> Earth over Heaven  
13 ☴ <-> 07 ☵ distance: 6 Heaven over Fire <-> Earth over Water  
14 ☳ <-> 08 ☴ distance: 6 Fire over Heaven <-> Water over Earth  
15 ☶ <-> 10 ☱ distance: 5 Earth over Mountain <-> Heaven over Lake  
16 ☱ <-> 09 ☲ distance: 7 Thunder over Earth <-> Wind over Heaven  
17 ☲ <-> 18 ☵ distance: 1 Lake over Thunder <-> Mountain over Wind  
18 ☵ <-> 17 ☲ distance: 1 Mountain over Wind <-> Lake over Thunder  
19 ☴ <-> 33 ☶ distance: 14 Earth over Lake <-> Heaven over Mountain  
20 ☲ <-> 34 ☵ distance: 14 Wind over Earth <-> Thunder over Heaven  
21 ☵ <-> 48 ☲ distance: 27 Fire over Thunder <-> Water over Wind

```
22 ䷮ <-> 47 ䷮ distance: 25 Mountain over Fire <-> Lake over Water
23 ䷮ <-> 43 ䷮ distance: 20 Mountain over Earth <-> Lake over Heaven
24 ䷮ <-> 44 ䷮ distance: 20 Earth over Thunder <-> Heaven over Wind
25 ䷮ <-> 46 ䷮ distance: 21 Heaven over Thunder <-> Earth over Wind
26 ䷮ <-> 45 ䷮ distance: 19 Mountain over Heaven <-> Lake over Earth
27 ䷮ <-> 28 ䷮ distance: 1 Mountain over Thunder <-> Lake over Wind
28 ䷮ <-> 27 ䷮ distance: 1 Lake over Wind <-> Mountain over Thunder
29 ䷮ <-> 30 ䷮ distance: 1 Water over Water <-> Fire over Fire
30 ䷮ <-> 29 ䷮ distance: 1 Fire over Fire <-> Water over Water
31 ䷮ <-> 41 ䷮ distance: 10 Lake over Mountain <-> Mountain over Lake
32 ䷮ <-> 42 ䷮ distance: 10 Thunder over Wind <-> Wind over Thunder
33 ䷮ <-> 19 ䷮ distance: 14 Heaven over Mountain <-> Earth over Lake
34 ䷮ <-> 20 ䷮ distance: 14 Thunder over Heaven <-> Wind over Earth
35 ䷮ <-> 05 ䷮ distance: 30 Fire over Earth <-> Water over Heaven
36 ䷮ <-> 06 ䷮ distance: 30 Earth over Fire <-> Heaven over Water
37 ䷮ <-> 40 ䷮ distance: 3 Wind over Fire <-> Thunder over Water
38 ䷮ <-> 39 ䷮ distance: 1 Fire over Lake <-> Water over Mountain
39 ䷮ <-> 38 ䷮ distance: 1 Water over Mountain <-> Fire over Lake
40 ䷮ <-> 37 ䷮ distance: 3 Thunder over Water <-> Wind over Fire
41 ䷮ <-> 31 ䷮ distance: 10 Mountain over Lake <-> Lake over Mountain
42 ䷮ <-> 32 ䷮ distance: 10 Wind over Thunder <-> Thunder over Wind
43 ䷮ <-> 23 ䷮ distance: 20 Lake over Heaven <-> Mountain over Earth
44 ䷮ <-> 24 ䷮ distance: 20 Heaven over Wind <-> Earth over Thunder
45 ䷮ <-> 26 ䷮ distance: 19 Lake over Earth <-> Mountain over Heaven
46 ䷮ <-> 25 ䷮ distance: 21 Earth over Wind <-> Heaven over Thunder
47 ䷮ <-> 22 ䷮ distance: 25 Lake over Water <-> Mountain over Fire
48 ䷮ <-> 21 ䷮ distance: 27 Water over Wind <-> Fire over Thunder
49 ䷮ <-> 04 ䷮ distance: 45 Lake over Fire <-> Mountain over Water
50 ䷮ <-> 03 ䷮ distance: 47 Fire over Wind <-> Water over Thunder
51 ䷮ <-> 57 ䷮ distance: 6 Thunder over Thunder <-> Wind over Wind
52 ䷮ <-> 58 ䷮ distance: 6 Mountain over Mountain <-> Lake over Lake
53 ䷮ <-> 54 ䷮ distance: 1 Wind over Mountain <-> Thunder over Lake
54 ䷮ <-> 53 ䷮ distance: 1 Thunder over Lake <-> Wind over Mountain
55 ䷮ <-> 59 ䷮ distance: 4 Thunder over Fire <-> Wind over Water
56 ䷮ <-> 60 ䷮ distance: 4 Fire over Mountain <-> Water over Lake
57 ䷮ <-> 51 ䷮ distance: 6 Wind over Wind <-> Thunder over Thunder
58 ䷮ <-> 52 ䷮ distance: 6 Lake over Lake <-> Mountain over Mountain
59 ䷮ <-> 55 ䷮ distance: 4 Wind over Water <-> Thunder over Fire
60 ䷮ <-> 56 ䷮ distance: 4 Water over Lake <-> Fire over Mountain
61 ䷮ <-> 62 ䷮ distance: 1 Wind over Lake <-> Thunder over Mountain
62 ䷮ <-> 61 ䷮ distance: 1 Thunder over Mountain <-> Wind over Lake
63 ䷮ <-> 64 ䷮ distance: 1 Water over Fire <-> Fire over Water
64 ䷮ <-> 63 ䷮ distance: 1 Fire over Water <-> Water over Fire
```

--- Complement distance statistics ---

Mean distance: 12.1

Median distance: 6

Min distance: 1 (adjacent = pair is an inverse pair)

Max distance: 47

--- Complement distance null model ---

Random mean complement distance (over 10000 shuffles): 21.6

King Wen mean complement distance: 12.1

King Wen percentile vs random: 0.0%

Complements are significantly closer together than chance would predict:  
the ordering keeps opposites unusually near one another.

## Palindrome analysis of the difference wave

---

Palindrome analysis of the difference wave

A palindrome reads the same forwards and backwards (like 2,4,6,4,2). Palindromic runs in the difference wave are mirror symmetry – sections where the pattern of change rises and falls in a balanced way. Longer palindromes are less likely to occur by chance, so counts and lengths are measured against null models below.

---

Found 27 palindromic subsequences (showing longest 20)

Length 7 at positions 11-17: [6,2,2,4,2,2,6] 

Length 7 at positions 19-25: [4,3,2,2,2,3,4] 

Length 7 at positions 39-45: [4,3,2,4,2,3,4] 



Length 5 at positions 12-16: [2,2,4,2,2]   
Length 5 at positions 20-24: [3,2,2,2,3]   
Length 5 at positions 40-44: [3,2,4,2,3]   
Length 5 at positions 43-47: [2,3,4,3,2]   
Length 5 at positions 45-49: [4,3,2,3,4]   
Length 5 at positions 55-59: [2,3,4,3,2]   
Length 3 at positions 03-05: [4,4,4]   
Length 3 at positions 07-09: [2,4,2]   
Length 3 at positions 08-10: [4,2,4]   
Length 3 at positions 13-15: [2,4,2]   
Length 3 at positions 18-20: [3,4,3]   
Length 3 at positions 21-23: [2,2,2]   
Length 3 at positions 26-28: [2,6,2]   
Length 3 at positions 27-29: [6,2,6]   
Length 3 at positions 30-32: [3,2,3]   
Length 3 at positions 33-35: [4,4,4]   
Length 3 at positions 35-37: [4,2,4] 

--- Palindrome count by length ---  
Length 7: 3 found  
Length 5: 6 found  
Length 3: 18 found

--- Palindrome null model ---  
How many palindromes of length >= 3 do random permutations produce?  
King Wen palindromes: 27, longest: 7  
Random mean palindromes: 22.4, mean longest: 7.0  
King Wen palindrome count percentile: 79.4%  
King Wen longest palindrome percentile: 40.3%

--- Pair-constrained palindrome null model ---  
The unconstrained comparison shuffles all 64 hexagrams freely. A fairer comparison preserves the pair structure: each pair of hexagrams stays adjacent, only the pair order and within-pair orientation are randomized.  
King Wen palindromes: 27, longest: 7  
Pair-constrained mean palindromes: 28.1, mean longest: 8.6  
King Wen palindrome count percentile (pair-constrained): 47.0%  
King Wen longest palindrome percentile (pair-constrained): 10.3%

## Upper Canon (1-30) vs. Lower Canon (31-64) comparison

---  
Upper Canon (1-30) vs. Lower Canon (31-64) comparison  
The I Ching is traditionally divided into two books: the Upper Canon (hexagrams 1-30) and Lower Canon (hexagrams 31-64). This division is ancient and may reflect different authorship, time periods, or thematic groupings. This section compares the mathematical properties of each half: do they have similar difference wave patterns? Similar yin/yang balance? Does the 'no 5-line transition' property hold in each half independently, or only in the whole?

| Metric             | Upper (1-30) | Lower (31-64) |
|--------------------|--------------|---------------|
| Hexagrams          | 30           | 34            |
| Transitions        | 29           | 33            |
| Mean line changes  | 3.38         | 3.33          |
| Yang lines (total) | 86           | 106           |
| Yin lines (total)  | 94           | 98            |

| Difference distribution: | Upper (1-30) | Lower (31-64) |
|--------------------------|--------------|---------------|
| 0 line changes           | 0            | 0             |
| 1 line changes           | 0            | 2             |
| 2 line changes           | 12           | 8             |
| 3 line changes           | 4            | 8             |
| 4 line changes           | 8            | 11            |
| 5 line changes           | 0            | 0             |
| 6 line changes           | 5            | 4             |

Upper Canon wave:   
Lower Canon wave: 

5-line transitions in Upper Canon: 0

5-line transitions in Lower Canon: 0

--- Canon split null model ---  
Permutation test: is the King Wen split at position 30 special, or would any random split of the 64-hexagram sequence show a similar gap in mean line-change differences between the two halves?  
King Wen |upper\_mean - lower\_mean|: 0.0460  
Random permutations with gap >= King Wen: 8707/10000  
King Wen gap percentile: 12.9%  
The King Wen canon split does not produce a statistically significant mean-difference gap. The observed difference between upper and lower halves is consistent with what random shuffles produce.

## Autocorrelation of the difference wave

---  
Autocorrelation of the difference wave  
Autocorrelation measures whether the wave is correlated with a shifted copy of itself. A peak at lag N means the pattern tends to repeat every N steps. A value near +1 means strong repetition, near -1 means strong alternation, and near 0 means no relationship. If the King Wen ordering contains hidden periodic structure, it will show up as peaks in this analysis.  
Caveat: with only N=63 data points, the 95% noise threshold is +/-0.25.  
Values within that band are indistinguishable from random noise.  
---  
Mean difference: 3.349  
Variance: 1.910  
95% noise threshold: +/-0.247 (values inside this are not significant)

| Lag | Autocorrelation | Sig? | Visualization |
|-----|-----------------|------|---------------|
| --- | -----           | ---- | -----         |
| 0   | 1.0000          |      | [ :   : #]    |
| 1   | -0.2469         |      | [ #   : ]     |
| 2   | -0.0098         |      | [ :   : ]     |
| 3   | -0.0491         |      | [ :   : ]     |
| 4   | -0.0447         |      | [ :   : ]     |
| 5   | -0.0457         |      | [ :   : ]     |
| 6   | -0.0218         |      | [ :   : ]     |
| 7   | -0.0644         |      | [ :   : ]     |
| 8   | 0.1204          |      | [ :  # : ]    |
| 9   | -0.1275         |      | [ : #  : ]    |
| 10  | 0.1375          |      | [ :   #: ]    |
| 11  | -0.0290         |      | [ :   : ]     |
| 12  | -0.0221         |      | [ :   : ]     |
| 13  | 0.0850          |      | [ :  # : ]    |
| 14  | -0.1596         |      | [ :#   : ]    |
| 15  | -0.0110         |      | [ :   : ]     |
| 16  | 0.1015          |      | [ :  # : ]    |
| 17  | -0.1514         |      | [ :#   : ]    |
| 18  | 0.0055          |      | [ :   : ]     |
| 19  | 0.0851          |      | [ :  # : ]    |
| 20  | -0.1403         |      | [ :#   : ]    |
| 21  | 0.0274          |      | [ :   : ]     |
| 22  | -0.1315         |      | [ : #  : ]    |
| 23  | -0.0137         |      | [ :   : ]     |
| 24  | 0.2263          |      | [ :   # ]     |
| 25  | -0.1595         |      | [ :#   : ]    |
| 26  | 0.1112          |      | [ :  # : ]    |
| 27  | -0.1109         |      | [ : #  : ]    |
| 28  | 0.1649          |      | [ :   #: ]    |
| 29  | 0.0093          |      | [ :   : ]     |
| 30  | -0.0969         |      | [ : #  : ]    |
| 31  | -0.0480         |      | [ :   : ]     |

Significant lags (outside noise band): 0/31  
(Expected false positives at 95% threshold with 31 lags: ~1.6)  
No significant autocorrelation detected – consistent with no hidden periodicity, but note that N=63 provides limited statistical power to detect weak periodicity.

# Spectral analysis (Discrete Fourier Transform)

---  
Spectral analysis (Discrete Fourier Transform)  
The DFT decomposes the difference wave into frequency components, like splitting white light into a rainbow. Each frequency represents a periodic pattern in the wave. A strong peak at frequency F means the wave has a repeating cycle every 63/F transitions. If the King Wen ordering has hidden periodic structure, the spectrum will show clear peaks rather than flat noise.  
Caveat: with only 63 samples, each of the 31 frequency bins has wide uncertainty. A flat spectrum may indicate no periodicity, or insufficient data.  
---  
Number of samples: 63  
Mean (DC component): 3.349  
White noise floor: 0.1741 (magnitudes near this are not significant)

| Freq | Period | Magnitude | Sig? | Spectrum |
|------|--------|-----------|------|----------|
| 1    | 63.0   | 0.0224    |      | ##       |
| 2    | 31.5   | 0.1823    |      | #####    |
| 3    | 21.0   | 0.0145    |      | #        |
| 4    | 15.8   | 0.1359    |      | #####    |
| 5    | 12.6   | 0.2021    |      | #####    |
| 6    | 10.5   | 0.1157    |      | #####    |
| 7    | 9.0    | 0.2399    |      | #####    |
| 8    | 7.9    | 0.1092    |      | #####    |
| 9    | 7.0    | 0.0553    |      | ####     |
| 10   | 6.3    | 0.1630    |      | #####    |
| 11   | 5.7    | 0.2371    |      | #####    |
| 12   | 5.2    | 0.0473    |      | ####     |
| 13   | 4.8    | 0.2326    |      | #####    |
| 14   | 4.5    | 0.0331    |      | ###      |
| 15   | 4.2    | 0.1145    |      | #####    |
| 16   | 3.9    | 0.2204    |      | #####    |
| 17   | 3.7    | 0.1451    |      | #####    |
| 18   | 3.5    | 0.1975    |      | #####    |
| 19   | 3.3    | 0.2197    |      | #####    |
| 20   | 3.1    | 0.2166    |      | #####    |
| 21   | 3.0    | 0.1611    |      | #####    |
| 22   | 2.9    | 0.0222    |      | ##       |
| 23   | 2.7    | 0.1210    |      | #####    |
| 24   | 2.6    | 0.4266    | *    | #####    |
| 25   | 2.5    | 0.0306    |      | ##       |
| 26   | 2.4    | 0.2297    |      | #####    |
| 27   | 2.3    | 0.0795    |      | #####    |
| 28   | 2.2    | 0.0639    |      | ####     |
| 29   | 2.2    | 0.2114    |      | #####    |
| 30   | 2.1    | 0.2129    |      | #####    |
| 31   | 2.0    | 0.2033    |      | #####    |

Frequencies above 2x noise floor: 1/31  
(The 2x threshold is ad hoc. A proper test would use Fisher's g-statistic or Bonferroni correction across frequency bins. With only 63 samples, even real periodicity may not rise above the noise floor.)

# Markov chain analysis

---  
Markov chain analysis  
If we treat each difference value as a 'state', we can ask: does the current difference predict the next one? In a random sequence, knowing that the last transition changed 2 lines tells you nothing about the next transition. But if certain patterns like '6 is always followed by 2' emerge, it reveals hidden rules in the King Wen ordering. The matrix shows the probability of each transition between difference values.  
---  
Transition probability matrix (row = from, column = to)  
-> 1    -> 2    -> 3    -> 4    -> 6  
1    0.00    0.00    0.00    0.00    1.00    (n=2)

|   |      |      |      |      |      |        |
|---|------|------|------|------|------|--------|
| 2 | 0.05 | 0.25 | 0.25 | 0.30 | 0.15 | (n=20) |
| 3 | 0.00 | 0.46 | 0.00 | 0.46 | 0.08 | (n=13) |
| 4 | 0.05 | 0.26 | 0.26 | 0.32 | 0.11 | (n=19) |
| 6 | 0.00 | 0.50 | 0.38 | 0.12 | 0.00 | (n=8)  |

```

--- Notable transition patterns ---
(Caveat: rows with small n are unreliable – patterns based on <5 observations
are anecdotes, not statistically meaningful.)
3 -> 2: 46% (6/13 times)
3 -> 4: 46% (6/13 times)

--- Permutation test ---
Is the transition matrix more concentrated than random orderings produce?
King Wen matrix concentration: 2.3303
Random orderings equally or more concentrated: 2157/5000 (43.1%)

```

## Gray code comparison

```

---
Gray code comparison
A Gray code is an ordering of binary values where each consecutive pair differs
by exactly 1 bit. For 6-bit hexagrams, a Gray code path would change exactly 1
line at every step – the smoothest possible sequence. The King Wen sequence does
NOT follow a Gray code (most transitions change 2-6 lines). This comparison
shows how much 'rougher' King Wen is than the theoretical minimum, suggesting
the ordering prioritizes something other than smooth transitions.
---

```

| Metric                     | Gray Code | King Wen |
|----------------------------|-----------|----------|
| Total path length          | 63        | 211      |
| Mean change per transition | 1.00      | 3.35     |
| Max single transition      | 1         | 6        |
| Min single transition      | 1         | 1        |
| King Wen / Gray Code ratio |           | 3.35x    |

  

| Difference distribution: | Gray Code | King Wen |
|--------------------------|-----------|----------|
| 1 line changes           | 63        | 2        |
| 2 line changes           | 0         | 20       |
| 3 line changes           | 0         | 13       |
| 4 line changes           | 0         | 19       |
| 6 line changes           | 0         | 9        |

  

```

Gray code wave:
King Wen wave:

```



```

--- Methodological note ---
This comparison is descriptive. The Gray code ratio (King Wen path / Gray
code path) is not a statistical test. For significance testing of King Wen's
path length against appropriate null models, see --path.

```

## Symmetry group analysis (XOR algebra)

```

---
Symmetry group analysis (XOR algebra)
The 64 hexagrams form a mathematical group under XOR: combining any two hexagrams
by toggling their differing lines always produces another valid hexagram. This is
like mixing colors – every combination is itself a color. Under this algebra,
certain sets of hexagrams form 'subgroups' (closed sets where combining any two
members stays in the set). This section checks whether the King Wen pairs, inverse
pairs, and the canonical divisions correspond to meaningful algebraic subgroups.
---
XOR identity element: 0b000000 (hexagram 2, all broken)

--- Fixed points under complement (XOR 0b111111) ---
Hexagrams that are their own complement do not exist (since XOR with
0b111111 always flips all bits, no 6-bit value equals its own complement).

--- King Wen pair XOR products ---

```

For each pair, XOR the two hexagrams to see what 'difference element' they produce.  
If many pairs produce the same XOR product, the pairing has algebraic regularity.

| XOR Product | Hexagram | Pair count                                           | Pairs |
|-------------|----------|------------------------------------------------------|-------|
| 001100 62 ䷋ | 4        | 9-10, 15-16, 21-22, 47-48                            |       |
| 010010 29 ䷌ | 4        | 7-8, 13-14, 31-32, 41-42                             |       |
| 011110 28 ䷍ | 4        | 25-26, 37-38, 39-40, 45-46                           |       |
| 100001 27 ䷎ | 4        | 23-24, 43-44, 55-56, 59-60                           |       |
| 101101 30 ䷏ | 4        | 5-6, 35-36, 51-52, 57-58                             |       |
| 110011 61 ䷐ | 4        | 3-4, 19-20, 33-34, 49-50                             |       |
| 111111 01 ䷑ | 8        | 1-2, 11-12, 17-18, 27-28, 29-30, 53-54, 61-62, 63-64 |       |

Unique XOR products: 7 (out of 32 pairs)  
If this number is small, the pairs share algebraic structure.

--- Inverse pair XOR closure test ---  
Inverse pair hexagrams (16 total): 01, 02, 11, 12, 17, 18, 27, 28, 29, 30, 53, 54, 61, 62, 63, 64  
Closed under XOR: Yes – forms a subgroup

## Alternative sequence comparison

---  
Alternative sequence comparison  
The King Wen ordering is not the only way to arrange 64 hexagrams. The Fu Xi (binary) sequence orders them by numerical value (0-63), which is mathematically natural but has no traditional significance. The Mawangdui sequence was found on silk manuscripts in a 168 BCE tomb and may represent an independent tradition. Comparing the same analyses across orderings reveals what is unique to King Wen.

| Metric             | King Wen | Fu Xi (binary) | Mawangdui |
|--------------------|----------|----------------|-----------|
| Total path length  | 211      | 120            | 141       |
| Mean change        | 3.35     | 1.90           | 2.24      |
| 1-line transitions | 2        | 32             | 21        |
| 2-line transitions | 20       | 16             | 10        |
| 3-line transitions | 13       | 8              | 29        |
| 4-line transitions | 19       | 4              | 2         |
| 5-line transitions | 0        | 2              | 1         |
| 6-line transitions | 9        | 1              | 0         |



--- What makes King Wen unique ---  
Zero 5-line transitions: King Wen=0, Fu Xi=2, Mawangdui=1  
Zero 0-line transitions: King Wen=0, Fu Xi=0, Mawangdui=0

## Windowed entropy analysis

---  
Windowed entropy analysis  
Instead of one entropy value for the whole wave, we slide a window across it and compute entropy locally. This reveals WHERE in the sequence structure concentrates (low entropy = predictable) vs. dissolves (high entropy = varied). Dips in the curve mark regions with repetitive transition patterns.

| Window size: 15 |         |               |
|-----------------|---------|---------------|
| Center          | Entropy | Visualization |
| 8               | 1.7056  | #####         |
| 9               | 1.5628  | #####         |
| 10              | 1.7056  | #####         |
| 11              | 1.8323  | #####         |
| 12              | 1.8323  | #####         |
| 13              | 1.8892  | #####         |
| 14              | 1.7968  | #####         |

```
15 1.7968 #####
16 1.7232 #####
17 1.8295 #####
18 1.8295 #####
19 1.6729 #####
20 1.8295 #####
21 1.8295 #####
22 1.8295 #####
23 1.8892 #####
24 1.8892 #####
25 1.8323 #####
26 1.8892 #####
27 1.8892 #####
28 1.8892 #####
29 1.8892 #####
30 1.9086 #####
31 1.9656 #####
32 1.8892 #####
33 1.9656 #####
34 1.9656 #####
35 1.8892 #####
36 1.8892 #####
37 1.8062 #####
38 1.7465 #####
39 1.7465 #####
40 1.7465 #####
41 1.8062 #####
42 1.8062 #####
43 1.8062 #####
44 1.7465 #####
45 2.0226 #####
46 2.0226 #####
47 2.0662 #####
48 2.0419 #####
49 2.0662 #####
50 2.0662 #####
51 2.0419 #####
52 2.0662 #####
53 2.1736 #####
54 2.2566 #####
55 2.2566 #####
56 2.2892 #####
```

Mean windowed entropy: 1.8956  
Min entropy: 1.5628 at position 9 (most structured)  
Max entropy: 2.2892 at position 56 (most varied)

Entropy spark: 

--- Methodological note ---  
Windowed entropy is an exploratory visualization, not a statistical test.  
Apparent patterns (dips and peaks) are expected from random variation in a short sequence. Without a null model, no significance can be assigned to specific regions. Interpret as descriptive, not inferential.

## Mutual information: upper vs. lower trigram transitions

---  
Mutual information: upper vs. lower trigram transitions  
When two consecutive hexagrams differ, do the upper and lower trigrams change independently, or is knowing that the upper trigram changed informative about whether the lower trigram also changed? Mutual information quantifies this:  
0 bits = completely independent (knowing one tells you nothing about the other)  
higher = correlated (they tend to change together or stay together)

---  
Joint distribution of trigram changes:

|                 | Lower unchanged | Lower changed |
|-----------------|-----------------|---------------|
| Upper unchanged | 0 ( 0.0%)       | 4 ( 6.3%)     |
| Upper changed   | 5 ( 7.9%)       | 54 ( 85.7%)   |

Mutual information: 0.0078 bits

Normalized MI: 0.0230 (0=independent, 1=perfectly correlated)

Mean MI of random permutations: 0.0203 bits  
King Wen percentile: 8.3%

--- Full 8-state trigram mutual information ---  
Using actual trigram identities (8 possible values each) rather than binary changed/unchanged, which preserves more information.  
MI(upper, lower) across all 64 hexagrams: 0.000000 bits  
(Expected: 0.0 – since all 64 combinations appear exactly once, the trigrams are perfectly independent in the static set.)  
Note: with all 64 upper/lower trigram combinations present exactly once, independence is expected by construction (complete Latin square).

# Yin-yang balance wave

---  
Yin-yang balance wave  
Each hexagram has 6 lines that are either yang (solid, 1) or yin (broken, 0).  
Note: since the King Wen sequence contains all 64 possible hexagrams exactly once, the total yin-yang count is necessarily balanced (192 each). This is a mathematical tautology, not a property of the ordering. However, the LOCAL balance as you move through the sequence IS a property of the ordering – it shows where yang-dominant or yin-dominant hexagrams cluster together.  
---  
Total yang lines: 192 / 384 (50.0%)  
Total yin lines: 192 / 384 (50.0%)  
Mean yang per hexagram: 3.00

| Running yang average (window=8): |      |        |               |   |    |
|----------------------------------|------|--------|---------------|---|----|
| Pos                              | Yang | RunAvg | Visualization |   |    |
| 1                                | 6    | 6.00   | [             |   | #] |
| 2                                | 0    | 3.00   | [             | # | ]  |
| 3                                | 2    | 2.67   | [             | # | ]  |
| 4                                | 2    | 2.50   | [             | # | ]  |
| 5                                | 4    | 2.80   | [             | # | ]  |
| 6                                | 4    | 3.00   | [             | # | ]  |
| 7                                | 1    | 2.71   | [             | # | ]  |
| 8                                | 1    | 2.50   | [             | # | ]  |
| 9                                | 5    | 2.38   | [             | # | ]  |
| 10                               | 5    | 3.00   | [             | # | ]  |
| 11                               | 3    | 3.12   | [             | # | ]  |
| 12                               | 3    | 3.25   | [             | # | ]  |
| 13                               | 5    | 3.38   | [             | # | ]  |
| 14                               | 5    | 3.50   | [             | # | ]  |
| 15                               | 1    | 3.50   | [             | # | ]  |
| 16                               | 1    | 3.50   | [             | # | ]  |
| 17                               | 3    | 3.25   | [             | # | ]  |
| 18                               | 3    | 3.00   | [             | # | ]  |
| 19                               | 2    | 2.88   | [             | # | ]  |
| 20                               | 2    | 2.75   | [             | # | ]  |
| 21                               | 3    | 2.50   | [             | # | ]  |
| 22                               | 3    | 2.25   | [             | # | ]  |
| 23                               | 1    | 2.25   | [             | # | ]  |
| 24                               | 1    | 2.25   | [             | # | ]  |
| 25                               | 4    | 2.38   | [             | # | ]  |
| 26                               | 4    | 2.50   | [             | # | ]  |
| 27                               | 2    | 2.50   | [             | # | ]  |
| 28                               | 4    | 2.75   | [             | # | ]  |
| 29                               | 2    | 2.62   | [             | # | ]  |
| 30                               | 4    | 2.75   | [             | # | ]  |
| 31                               | 3    | 3.00   | [             | # | ]  |
| 32                               | 3    | 3.25   | [             | # | ]  |
| 33                               | 4    | 3.25   | [             | # | ]  |
| 34                               | 4    | 3.25   | [             | # | ]  |
| 35                               | 2    | 3.25   | [             | # | ]  |
| 36                               | 2    | 3.00   | [             | # | ]  |
| 37                               | 4    | 3.25   | [             | # | ]  |
| 38                               | 4    | 3.25   | [             | # | ]  |
| 39                               | 2    | 3.12   | [             | # | ]  |
| 40                               | 2    | 3.00   | [             | # | ]  |

|    |   |      |   |   |   |
|----|---|------|---|---|---|
| 41 | 3 | 2.88 | [ | # | ] |
| 42 | 3 | 2.75 | [ | # | ] |
| 43 | 5 | 3.12 | [ | # | ] |
| 44 | 5 | 3.50 | [ | # | ] |
| 45 | 2 | 3.25 | [ | # | ] |
| 46 | 2 | 3.00 | [ | # | ] |
| 47 | 3 | 3.12 | [ | # | ] |
| 48 | 3 | 3.25 | [ | # | ] |
| 49 | 4 | 3.38 | [ | # | ] |
| 50 | 4 | 3.50 | [ | # | ] |
| 51 | 2 | 3.12 | [ | # | ] |
| 52 | 2 | 2.75 | [ | # | ] |
| 53 | 3 | 2.88 | [ | # | ] |
| 54 | 3 | 3.00 | [ | # | ] |
| 55 | 3 | 3.00 | [ | # | ] |
| 56 | 3 | 3.00 | [ | # | ] |
| 57 | 4 | 3.00 | [ | # | ] |
| 58 | 4 | 3.00 | [ | # | ] |
| 59 | 3 | 3.12 | [ | # | ] |
| 60 | 3 | 3.25 | [ | # | ] |
| 61 | 4 | 3.38 | [ | # | ] |
| 62 | 2 | 3.25 | [ | # | ] |
| 63 | 3 | 3.25 | [ | # | ] |
| 64 | 3 | 3.25 | [ | # | ] |

Yang count spark: 

--- Yang/Yin dominance runs ---  
Positions 02-04: 3 consecutive yin-dominant hexagrams

## Odd-vs-even transition parity (McKenna 25/75)

---  
Odd-vs-even transition parity (McKenna's 25/75 observation)  
---  
Each transition  $d(s_i, s_{i+1})$  is between 1 and 6 line changes.  
Theorem 1 says within-pair distances are always even (in  $\{2,4,6\}$ ),  
so all 32 within-pair transitions are forced even by C1. The 31  
between-pair transitions can be any non-5 value.

McKenna (The Invisible Landscape, 1975) noted that ~25% of King Wen's  
transitions are odd-distance and ~75% are even-distance. We report both  
the LINEAR reading (the standard 63-transition difference wave) and the  
CIRCULAR reading (64 transitions, including the wrap-around  $s_{63} \rightarrow s_0$ ).

Last hexagram (position 64): #42 (101010)  
First hexagram (position 1): #63 (111111)  
Wrap-around distance:  $\text{hamming}(\#42, \#63) = 3$  (odd)

| Mode                | Total | Odd | %Odd     | Even | %Even    |
|---------------------|-------|-----|----------|------|----------|
| LINEAR (63 trans)   | 63    | 15  | 23.8095% | 48   | 76.1905% |
| CIRCULAR (64 trans) | 64    | 16  | 25.0000% | 48   | 75.0000% |

CIRCULAR parity is EXACTLY 25.0000% / 75.0000% – McKenna's 25/75 holds exactly  
when the wrap-around transition is included. Without wrap-around, the linear  
split is 23.8095% / 76.1905% – approximately but not exactly 25/75.

--- Structural note ---  
All 32 within-pair transitions are forced even by C1 + Theorem 1.  
So the parity split is determined entirely by the 31 (or 32 with wrap)  
between-pair transitions:  
Between-pair transitions: 15 odd, 16 even  
Within-pair transitions: 0 odd (forced), 32 even (forced)  
Plus wrap-around: 1 odd

--- 'C8' candidate: is the wrap-around parity a structural constraint? ---  
The spec's C1-C7 do not constrain the wrap-around  $s_{63} \rightarrow s_0$ . So if  
McKenna's exact 25/75 reflects a real constraint on the sequence rather  
than a numerical coincidence, it would imply a new constraint:  
C8 (weak):  $\text{hamming}(s_{63}, s_0)$  is odd (admits  $\{1, 3, 5\}$ ; C2 may extend to 64th)  
C8 (strict):  $\text{hamming}(s_{63}, s_0) = 3$  (King Wen's exact value)



Restrictiveness: the prior for wrap-around distance among {1,2,3,4,6} (5 forbidden by C2 if extended, 0 impossible for a permutation) is uniform → P(odd) ≈ 40%, so the weak form eliminates ~60% of completions; the strict form eliminates ~80%.

Whether C8 is a real constraint or a numerical coincidence is empirical: a Monte Carlo sample of sequences satisfying C1-C7 would tell us how non-uniform the wrap-around distribution actually is among King Wen-eligible orderings. Today this is NOT in the formal spec – it remains an observation, not a constraint.

## Hexagram neighborhoods

|                                                                                                                                                                                                  |                               |                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------|
| ---                                                                                                                                                                                              |                               |                                    |
| Hexagram neighborhoods                                                                                                                                                                           |                               |                                    |
| For each hexagram, which others are 'nearby' (differ by only 1 or 2 lines)?                                                                                                                      |                               |                                    |
| Dense neighborhoods mean a hexagram has many close relatives; sparse ones are more isolated. This also shows whether King Wen places neighbors near each other in the sequence or scatters them. |                               |                                    |
| ---                                                                                                                                                                                              |                               |                                    |
| 01 ☰ Heaven over Heaven                                                                                                                                                                          | d=1: [09, 10, 13, 14, 43, 44] | (6 neighbors, mean seq dist: 21.2) |
| 02 ☷ Earth over Earth                                                                                                                                                                            | d=1: [07, 08, 15, 16, 23, 24] | (6 neighbors, mean seq dist: 13.5) |
| 03 ☵ Water over Thunder                                                                                                                                                                          | d=1: [08, 17, 24, 42, 60, 63] | (6 neighbors, mean seq dist: 32.7) |
| 04 ☶ Mountain over Water                                                                                                                                                                         | d=1: [07, 18, 23, 41, 59, 64] | (6 neighbors, mean seq dist: 31.3) |
| 05 ☵ Water over Heaven                                                                                                                                                                           | d=1: [09, 11, 43, 48, 60, 63] | (6 neighbors, mean seq dist: 34.0) |
| 06 ☰ Heaven over Water                                                                                                                                                                           | d=1: [10, 12, 44, 47, 59, 64] | (6 neighbors, mean seq dist: 33.3) |
| 07 ☷ Earth over Water                                                                                                                                                                            | d=1: [02, 04, 19, 29, 40, 46] | (6 neighbors, mean seq dist: 19.0) |
| 08 ☵ Water over Earth                                                                                                                                                                            | d=1: [02, 03, 20, 29, 39, 45] | (6 neighbors, mean seq dist: 18.7) |
| 09 ☰ Wind over Heaven                                                                                                                                                                            | d=1: [01, 05, 26, 37, 57, 61] | (6 neighbors, mean seq dist: 26.2) |
| 10 ☷ Heaven over Lake                                                                                                                                                                            | d=1: [01, 06, 25, 38, 58, 61] | (6 neighbors, mean seq dist: 25.8) |
| 11 ☷ Earth over Heaven                                                                                                                                                                           | d=1: [05, 19, 26, 34, 36, 46] | (6 neighbors, mean seq dist: 18.7) |
| 12 ☰ Heaven over Earth                                                                                                                                                                           | d=1: [06, 20, 25, 33, 35, 45] | (6 neighbors, mean seq dist: 17.3) |
| 13 ☷ Heaven over Fire                                                                                                                                                                            | d=1: [01, 25, 30, 33, 37, 49] | (6 neighbors, mean seq dist: 20.2) |
| 14 ☰ Fire over Heaven                                                                                                                                                                            | d=1: [01, 26, 30, 34, 38, 50] | (6 neighbors, mean seq dist: 20.2) |
| 15 ☷ Earth over Mountain                                                                                                                                                                         | d=1: [02, 36, 39, 46, 52, 62] | (6 neighbors, mean seq dist: 28.8) |
| 16 ☵ Thunder over Earth                                                                                                                                                                          | d=1: [02, 35, 40, 45, 51, 62] | (6 neighbors, mean seq dist: 27.8) |
| 17 ☵ Lake over Thunder                                                                                                                                                                           | d=1: [03, 25, 45, 49, 51, 58] | (6 neighbors, mean seq dist: 26.2) |
| 18 ☶ Mountain over Wind                                                                                                                                                                          | d=1: [04, 26, 46, 50, 52, 57] | (6 neighbors, mean seq dist: 25.8) |
| 19 ☷ Earth over Lake                                                                                                                                                                             | d=1: [07, 11, 24, 41, 54, 60] | (6 neighbors, mean seq dist: 20.5) |
| 20 ☵ Wind over Earth                                                                                                                                                                             | d=1: [08, 12, 23, 42, 53, 59] | (6 neighbors, mean seq dist: 19.5) |
| 21 ☰ Fire over Thunder                                                                                                                                                                           | d=1: [25, 27, 30, 35, 38, 51] | (6 neighbors, mean seq dist: 13.3) |
| 22 ☶ Mountain over Fire                                                                                                                                                                          | d=1: [26, 27, 30, 36, 37, 52] | (6 neighbors, mean seq dist: 12.7) |
| 23 ☷ Mountain over Earth                                                                                                                                                                         | d=1: [02, 04, 20, 27, 35, 52] | (6 neighbors, mean seq dist: 14.7) |
| 24 ☷ Earth over Thunder                                                                                                                                                                          | d=1: [02, 03, 19, 27, 36, 51] | (6 neighbors, mean seq dist: 15.0) |
| 25 ☰ Heaven over Thunder                                                                                                                                                                         | d=1: [10, 12, 13, 17, 21, 42] | (6 neighbors, mean seq dist: 11.5) |
| 26 ☶ Mountain over Heaven                                                                                                                                                                        | d=1: [09, 11, 14, 18, 22, 41] | (6 neighbors, mean seq dist: 11.8) |
| 27 ☶ Mountain over Thunder                                                                                                                                                                       | d=1: [21, 22, 23, 24, 41, 42] | (6 neighbors, mean seq dist: 7.8)  |
| 28 ☰ Lake over Wind                                                                                                                                                                              | d=1: [31, 32, 43, 44, 47, 48] | (6 neighbors, mean seq dist: 12.8) |
| 29 ☵ Water over Water                                                                                                                                                                            | d=1: [07, 08, 47, 48, 59, 60] | (6 neighbors, mean seq dist: 23.5) |
| 30 ☰ Fire over Fire                                                                                                                                                                              | d=1: [13, 14, 21, 22, 55, 56] | (6 neighbors, mean seq dist: 16.8) |
| 31 ☰ Lake over Mountain                                                                                                                                                                          | d=1: [28, 33, 39, 45, 49, 62] | (6 neighbors, mean seq dist: 12.7) |
| 32 ☵ Thunder over Wind                                                                                                                                                                           | d=1: [28, 34, 40, 46, 50, 62] | (6 neighbors, mean seq dist: 12.7) |
| 33 ☰ Heaven over Mountain                                                                                                                                                                        | d=1: [12, 13, 31, 44, 53, 56] | (6 neighbors, mean seq dist: 16.2) |
| 34 ☷ Thunder over Heaven                                                                                                                                                                         | d=1: [11, 14, 32, 43, 54, 55] | (6 neighbors, mean seq dist: 15.8) |
| 35 ☰ Fire over Earth                                                                                                                                                                             | d=1: [12, 16, 21, 23, 56, 64] | (6 neighbors, mean seq dist: 19.7) |
| 36 ☷ Earth over Fire                                                                                                                                                                             | d=1: [11, 15, 22, 24, 55, 63] | (6 neighbors, mean seq dist: 19.7) |
| 37 ☰ Wind over Fire                                                                                                                                                                              | d=1: [09, 13, 22, 42, 53, 63] | (6 neighbors, mean seq dist: 19.0) |
| 38 ☰ Fire over Lake                                                                                                                                                                              | d=1: [10, 14, 21, 41, 54, 64] | (6 neighbors, mean seq dist: 19.0) |
| 39 ☵ Water over Mountain                                                                                                                                                                         | d=1: [08, 15, 31, 48, 53, 63] | (6 neighbors, mean seq dist: 18.3) |
| 40 ☷ Thunder over Water                                                                                                                                                                          | d=1: [07, 16, 32, 47, 54, 64] | (6 neighbors, mean seq dist: 18.3) |
| 41 ☶ Mountain over Lake                                                                                                                                                                          | d=1: [04, 19, 26, 27, 38, 61] | (6 neighbors, mean seq dist: 18.5) |
| 42 ☵ Wind over Thunder                                                                                                                                                                           | d=1: [03, 20, 25, 27, 37, 61] | (6 neighbors, mean seq dist: 19.5) |
| 43 ☰ Lake over Heaven                                                                                                                                                                            | d=1: [01, 05, 28, 34, 49, 58] | (6 neighbors, mean seq dist: 20.8) |
| 44 ☰ Heaven over Wind                                                                                                                                                                            | d=1: [01, 06, 28, 33, 50, 57] | (6 neighbors, mean seq dist: 21.2) |
| 45 ☰ Lake over Earth                                                                                                                                                                             | d=1: [08, 12, 16, 17, 31, 47] | (6 neighbors, mean seq dist: 23.8) |
| 46 ☷ Earth over Wind                                                                                                                                                                             | d=1: [07, 11, 15, 18, 32, 48] | (6 neighbors, mean seq dist: 24.8) |
| 47 ☰ Lake over Water                                                                                                                                                                             | d=1: [06, 28, 29, 40, 45, 58] | (6 neighbors, mean seq dist: 16.3) |
| 48 ☵ Water over Wind                                                                                                                                                                             | d=1: [05, 28, 29, 39, 46, 57] | (6 neighbors, mean seq dist: 17.0) |
| 49 ☰ Lake over Fire                                                                                                                                                                              | d=1: [13, 17, 31, 43, 55, 63] | (6 neighbors, mean seq dist: 18.7) |
| 50 ☰ Fire over Wind                                                                                                                                                                              | d=1: [14, 18, 32, 44, 56, 64] | (6 neighbors, mean seq dist: 18.7) |
| 51 ☷ Thunder over Thunder                                                                                                                                                                        | d=1: [16, 17, 21, 24, 54, 55] | (6 neighbors, mean seq dist: 22.2) |
| 52 ☶ Mountain over Mountain                                                                                                                                                                      | d=1: [15, 18, 22, 23, 53, 56] | (6 neighbors, mean seq dist: 22.5) |

```
53 ䷗ Wind over Mountain      d=1: [20, 33, 37, 39, 52, 57] (6 neighbors, mean seq dist: 14.7)
54 ䷙ Thunder over Lake       d=1: [19, 34, 38, 40, 51, 58] (6 neighbors, mean seq dist: 15.3)
55 ䷙ Thunder over Fire       d=1: [30, 34, 36, 49, 51, 62] (6 neighbors, mean seq dist: 13.7)
56 ䷗ Fire over Mountain      d=1: [30, 33, 35, 50, 52, 62] (6 neighbors, mean seq dist: 14.3)
57 ䷗ Wind over Wind          d=1: [09, 18, 44, 48, 53, 59] (6 neighbors, mean seq dist: 19.2)
58 ䷙ Lake over Lake          d=1: [10, 17, 43, 47, 54, 60] (6 neighbors, mean seq dist: 20.2)
59 ䷗ Wind over Water          d=1: [04, 06, 20, 29, 57, 61] (6 neighbors, mean seq dist: 30.2)
60 ䷙ Water over Lake          d=1: [03, 05, 19, 29, 58, 61] (6 neighbors, mean seq dist: 31.2)
61 ䷗ Wind over Lake           d=1: [09, 10, 41, 42, 59, 60] (6 neighbors, mean seq dist: 24.2)
62 ䷗ Thunder over Mountain    d=1: [15, 16, 31, 32, 55, 56] (6 neighbors, mean seq dist: 27.8)
63 ䷙ Water over Fire          d=1: [03, 05, 36, 37, 39, 49] (6 neighbors, mean seq dist: 34.8)
64 ䷗ Fire over Water          d=1: [04, 06, 35, 38, 40, 50] (6 neighbors, mean seq dist: 35.2)
```

--- Neighborhood clustering in King Wen sequence ---

Mean sequence distance between Hamming-1 neighbors: 20.6

Mean sequence distance between any two hexagrams: 21.7

Hamming-1 neighbors are closer than average in the sequence (clustered).

--- Neighborhood clustering null model ---

Shuffling binary hexagrams 10,000 times to build a null distribution  
of mean sequence distance between Hamming-1 neighbors.

Null distribution: min=17.6, mean=21.7, max=25.2

King Wen observed mean: 20.6

Percentile: 11.9% (proportion of shuffles with mean <= King Wen's)

King Wen's neighborhood clustering is within the range expected by chance.

## Recurrence plot

---

Recurrence plot

A recurrence plot shows where the difference wave repeats itself. Each cell (i,j)

is marked if the difference at position i equals the difference at position j.

Diagonal lines indicate repeated sequences; vertical/horizontal lines indicate

values that persist. Clusters of marks reveal recurring local patterns.

---

```
          1         2         3         4         5         6
12345678901234567890123456789012345678901234567890123
1  +.....#.....#.....#.....#.....#.....#.....#
2  .+....#.#.##.##.###.##.##.##.##.##.##.##.##.##.##
3  ..+###.##.##.##.##.##.##.##.##.##.##.##.##.##.##
4  .#+#.##.##.##.##.##.##.##.##.##.##.##.##.##.##
5  .##+.##.##.##.##.##.##.##.##.##.##.##.##.##.##
6  .....+.....#.#.##.##.##.##.##.##.##.##.##.##.##
7  .#....+.##.##.##.###.##.##.##.##.##.##.##.##.##
8  .###.##+.##.##.##.##.##.##.##.##.##.##.##.##
9  .#....#+.##.##.##.###.##.##.##.##.##.##.##.##.##
10 .###.##+.##.##.##.##.##.##.##.##.##.##.##.##
11 #.....+.##.##.##.##.##.##.##.##.##.##.##.##.##
12 .#....#.#.##+.##.##.##.##.##.##.##.##.##.##.##
13 .#....#.#.##+.##.##.##.##.##.##.##.##.##.##.##
14 .###.##.##.##.##.##.##.##.##.##.##.##.##.##.##
15 .#....#.#.##+.##.##.##.##.##.##.##.##.##.##.##
16 .#....#.#.##.##+.##.##.##.##.##.##.##.##.##.##
17 #.....#.....+.##.##.##.##.##.##.##.##.##.##
18 .....#.....+.##.##.##.##.##.##.##.##.##.##.##
19 .###.##.##.##.##.##.##.##.##.##.##.##.##.##.##
20 .....#.....#.....+.##.##.##.##.##.##.##.##.##
21 .#....#.#.##.##.##.##.##.##.##.##.##.##.##.##
22 .#....#.#.##.##.##.##.##.##.##.##.##.##.##.##
23 .#....#.#.##.##.##.##.##.##.##.##.##.##.##.##
24 .....#.....#.....+.##.##.##.##.##.##.##.##.##
25 .###.##.##.##.##.##.##.##.##.##.##.##.##.##.##
26 .#....#.#.##.##.##.##.##.##.##.##.##.##.##.##
27 #.....#.....#.....+.##.##.##.##.##.##.##.##
28 .#....#.#.##.##.##.##.##.##.##.##.##.##.##.##
29 #.....#.....#.....#.....+.##.##.##.##.##.##
30 .....#.....#.....#.....+.##.##.##.##.##.##.##
31 .#....#.#.##.##.##.##.##.##.##.##.##.##.##.##
32 .....#.....#.....#.....#.....+.##.##.##.##.##
33 .###.##.##.##.##.##.##.##.##.##.##.##.##.##.##
34 .###.##.##.##.##.##.##.##.##.##.##.##.##.##.##
```

```

--- Recurrence rate null model ---
Theoretical expected recurrence rate (sum of p_i^2): 25.6%

King Wen recurrence rate:          24.4%
Mean random recurrence rate:       23.2%
King Wen percentile vs random:    72.9%

```

DNA codon mapping

Both the I Ching (64 hexagrams) and the genetic code (64 codons) are systems of 64 elements built from binary-like pairs: yin/yang for hexagrams, and the base pairs A-T and G-C for DNA. Each codon is a sequence of 3 bases from the set {A, C, G, U}, giving  $4^3 = 64$  possibilities. While this is a numerical coincidence (both are  $2^6 = 64$ ), comparing their structural properties is instructive about combinatorial systems in general.

Caveat: the mapping of bit pairs to DNA bases (00=U, 01=C, 10=A, 11=G) is one of 24 possible assignments. Different mappings produce different codon assignments and different degeneracy statistics. No mapping is privileged by either system.

This analysis is illustrative, not evidence of a biological connection.

| Pos | Hex | Binary | Codon | Amino Acid | Name                |
|-----|-----|--------|-------|------------|---------------------|
| 01  | ☰   | 111111 | GGG   | Gly        | Heaven over Heaven  |
| 02  | ☷   | 000000 | UUU   | Phe        | Earth over Earth    |
| 03  | ☳   | 010001 | CUC   | Leu        | Water over Thunder  |
| 04  | ☱   | 100010 | AUA   | Ile        | Mountain over Water |
| 05  | ☴   | 010111 | CCG   | Pro        | Water over Heaven   |
| 06  | ☵   | 111010 | GAA   | Glu        | Heaven over Water   |
| 07  | ☶   | 000010 | UUA   | Leu        | Earth over Water    |
| 08  | ☲   | 010000 | CUU   | Leu        | Water over Earth    |
| 09  | ☴   | 110111 | GCG   | Ala        | Wind over Heaven    |
| 10  | ☳   | 111011 | GAG   | Glu        | Heaven over Lake    |
| 11  | ☱   | 000111 | UCG   | Ser        | Earth over Heaven   |
| 12  | ☲   | 111000 | GAU   | Asp        | Heaven over Earth   |
| 13  | ☵   | 111101 | GGC   | Gly        | Heaven over Fire    |
| 14  | ☴   | 101111 | AGG   | Arg        | Fire over Heaven    |
| 15  | ☶   | 000100 | UCU   | Ser        | Earth over Mountain |

|    |  |   |  |        |  |     |  |      |  |                        |
|----|--|---|--|--------|--|-----|--|------|--|------------------------|
| 16 |  | ䷀ |  | 001000 |  | UAU |  | Tyr  |  | Thunder over Earth     |
| 17 |  | ䷁ |  | 011001 |  | CAC |  | His  |  | Lake over Thunder      |
| 18 |  | ䷂ |  | 100110 |  | ACA |  | Thr  |  | Mountain over Wind     |
| 19 |  | ䷃ |  | 000011 |  | UUG |  | Leu  |  | Earth over Lake        |
| 20 |  | ䷄ |  | 110000 |  | GUU |  | Val  |  | Wind over Earth        |
| 21 |  | ䷅ |  | 101001 |  | AAC |  | Asn  |  | Fire over Thunder      |
| 22 |  | ䷆ |  | 100101 |  | ACC |  | Thr  |  | Mountain over Fire     |
| 23 |  | ䷇ |  | 100000 |  | AUU |  | Ile  |  | Mountain over Earth    |
| 24 |  | ䷈ |  | 000001 |  | UUC |  | Phe  |  | Earth over Thunder     |
| 25 |  | ䷉ |  | 111001 |  | GAC |  | Asp  |  | Heaven over Thunder    |
| 26 |  | ䷊ |  | 100111 |  | ACG |  | Thr  |  | Mountain over Heaven   |
| 27 |  | ䷋ |  | 100001 |  | AUC |  | Ile  |  | Mountain over Thunder  |
| 28 |  | ䷌ |  | 011110 |  | CGA |  | Arg  |  | Lake over Wind         |
| 29 |  | ䷍ |  | 010010 |  | CUA |  | Leu  |  | Water over Water       |
| 30 |  | ䷎ |  | 101101 |  | AGC |  | Ser  |  | Fire over Fire         |
| 31 |  | ䷏ |  | 011100 |  | CGU |  | Arg  |  | Lake over Mountain     |
| 32 |  | ䷐ |  | 001110 |  | UGA |  | Stop |  | Thunder over Wind      |
| 33 |  | ䷑ |  | 111100 |  | GGU |  | Gly  |  | Heaven over Mountain   |
| 34 |  | ䷒ |  | 001111 |  | UGG |  | Trp  |  | Thunder over Heaven    |
| 35 |  | ䷓ |  | 101000 |  | AAU |  | Asn  |  | Fire over Earth        |
| 36 |  | ䷔ |  | 000101 |  | UCC |  | Ser  |  | Earth over Fire        |
| 37 |  | ䷕ |  | 110101 |  | GCC |  | Ala  |  | Wind over Fire         |
| 38 |  | ䷖ |  | 101011 |  | AAG |  | Lys  |  | Fire over Lake         |
| 39 |  | ䷗ |  | 010100 |  | CCU |  | Pro  |  | Water over Mountain    |
| 40 |  | ䷘ |  | 001010 |  | UAA |  | Stop |  | Thunder over Water     |
| 41 |  | ䷙ |  | 100011 |  | AUG |  | Met  |  | Mountain over Lake     |
| 42 |  | ䷚ |  | 110001 |  | GUC |  | Val  |  | Wind over Thunder      |
| 43 |  | ䷛ |  | 011111 |  | CGG |  | Arg  |  | Lake over Heaven       |
| 44 |  | ䷜ |  | 111110 |  | GGA |  | Gly  |  | Heaven over Wind       |
| 45 |  | ䷝ |  | 011000 |  | CAU |  | His  |  | Lake over Earth        |
| 46 |  | ䷞ |  | 000110 |  | UCA |  | Ser  |  | Earth over Wind        |
| 47 |  | ䷟ |  | 011010 |  | CAA |  | Gln  |  | Lake over Water        |
| 48 |  | ䷠ |  | 010110 |  | CCA |  | Pro  |  | Water over Wind        |
| 49 |  | ䷡ |  | 011101 |  | CGC |  | Arg  |  | Lake over Fire         |
| 50 |  | ䷢ |  | 101110 |  | AGA |  | Arg  |  | Fire over Wind         |
| 51 |  | ䷣ |  | 001001 |  | UAC |  | Tyr  |  | Thunder over Thunder   |
| 52 |  | ䷤ |  | 100100 |  | ACU |  | Thr  |  | Mountain over Mountain |
| 53 |  | ䷥ |  | 110100 |  | GCU |  | Ala  |  | Wind over Mountain     |
| 54 |  | ䷦ |  | 001011 |  | UAG |  | Stop |  | Thunder over Lake      |
| 55 |  | ䷧ |  | 001101 |  | UGC |  | Cys  |  | Thunder over Fire      |
| 56 |  | ䷨ |  | 101100 |  | AGU |  | Ser  |  | Fire over Mountain     |
| 57 |  | ䷩ |  | 110110 |  | GCA |  | Ala  |  | Wind over Wind         |
| 58 |  | ䷪ |  | 011011 |  | CAG |  | Gln  |  | Lake over Lake         |
| 59 |  | ䷫ |  | 110010 |  | GUA |  | Val  |  | Wind over Water        |
| 60 |  | ䷬ |  | 010011 |  | CUG |  | Leu  |  | Water over Lake        |
| 61 |  | ䷭ |  | 110011 |  | GUG |  | Val  |  | Wind over Lake         |
| 62 |  | ䷮ |  | 001100 |  | UGU |  | Cys  |  | Thunder over Mountain  |
| 63 |  | ䷯ |  | 010101 |  | CCC |  | Pro  |  | Water over Fire        |
| 64 |  | ䷰ |  | 101010 |  | AAA |  | Lys  |  | Fire over Water        |

Unique amino acids in King Wen order: 21  
Total possible amino acids + stop: 21

--- Degeneracy comparison ---

In DNA, many single-base changes preserve the amino acid (the code is 'degenerate').  
Do single-line hexagram changes preserve the mapped amino acid?  
Single-line changes that preserve amino acid: 100/384 (26.0%)

## Shannon entropy analysis

---

Shannon entropy analysis

Entropy measures disorder: high entropy means the difference values are spread evenly across all possibilities (random-looking), low entropy means certain values dominate (structured). We compare the King Wen wave's entropy against thousands of random permutations of the same 64 hexagrams. If King Wen's entropy is unusually low, the sequence is more structured than random chance would produce.

---

King Wen difference wave entropy: 2.0759 bits  
Maximum entropy (all 7 values): 2.8074 bits  
Maximum entropy (5 observed values): 2.3219 bits

Mean entropy of random permutations: 2.1927 bits  
Min random entropy observed: 1.6995 bits  
Max random entropy observed: 2.5081 bits  
King Wen percentile: 12.6% (lower = more structured)  
Effect size (Cohen's d): -1.14 (negative = more structured than random)

--- Entropy conditioned on pair constraint ---  
The unconstrained comparison above may be misleading: the pair structure itself constrains the entropy. How does King Wen compare against random orderings that also satisfy the pair constraint?  
Mean pair-constrained entropy: 2.2344 bits  
King Wen percentile (pair-constrained): 5.9%  
(Similar to unconstrained percentile of 12.6%.)

--- Distribution comparison ---

| Value | King Wen | Expected (random avg) |
|-------|----------|-----------------------|
| 0     | 0        | 0.0                   |
| 1     | 2        | 6.0                   |
| 2     | 20       | 15.2                  |
| 3     | 13       | 20.0                  |
| 4     | 19       | 14.8                  |
| 5     | 0        | 6.0                   |
| 6     | 9        | 1.0                   |

## Path analysis (graph theory)

---  
Path analysis (graph theory)  
Imagine the 64 hexagrams as cities on a map, where the 'distance' between any two is the number of lines that differ. The King Wen sequence is a route that visits all 64 cities. Is it a short, efficient route (nearby hexagrams placed next to each other), or a long, wandering one? We compare its total distance against random routes and a greedy 'always go to the nearest unvisited city' algorithm. This reveals whether the King Wen ordering was optimized for smooth transitions or had other priorities.  
---  
King Wen total path length: 211 (sum of all line changes)  
Greedy nearest-neighbor: 75  
Mean random path length: 192.1  
Min random observed: 161  
Max random observed: 227  
King Wen percentile: 97.6% (lower = shorter path)  
Effect size (Cohen's d): +2.03

--- Theoretical bounds ---  
Minimum possible (63 transitions of 1): 63  
Maximum possible (63 transitions of 6): 378  
King Wen uses 55.8% of maximum path length

--- Pair-constrained comparison ---  
The above compares against fully random orderings. A fairer comparison is against random orderings that also preserve the pair structure (each pair of hexagrams stays adjacent). This is the right null model for asking whether King Wen's path length is unusual GIVEN its pair constraint.  
Mean pair-constrained path length: 214.1  
Min pair-constrained observed: 191  
Max pair-constrained observed: 239  
King Wen percentile (pair-constrained): 28.2%  
Effect size (Cohen's d, pair-constrained): -0.48

## Constraint satisfaction analysis

---  
Constraint satisfaction analysis  
The King Wen sequence satisfies several constraints simultaneously:

1. All 32 consecutive pairs are either reverse or inverse (never unrelated)
2. No consecutive hexagrams differ by exactly 5 lines
3. No consecutive hexagrams are identical (0-line transitions)

How rare is it for a random ordering to satisfy all these at once? We test random permutations against each constraint individually and combined.

---

Results from 10,000 random permutations:

```
All pairs reverse/inverse:      0 (0.000%)
No 5-line transitions:          22 (0.22%)
Both constraints together:      0 (0.0000%)
No random permutation satisfied both constraints.
Statistical note: 0/10,000 gives a 95% upper bound of <0.0300%
(i.e., the true rate is less than ~1 in 3,333, but we cannot
distinguish 'impossible' from 'extremely rare' with this sample size).
```

--- Conditional probability: P(no-5 | pair constraint) ---

The pair structure constrains transitions within pairs (always even or 6), so 5-line transitions can only occur at the 31 between-pair boundaries.

How often do pair-constrained orderings also avoid 5-line transitions?

```
Pair-constrained trials: 100,000
Also satisfy no-5:        4,294 (4.29%)
Approximately 1 in 23 pair-constrained orderings avoid 5-line transitions.
The no-5 property is uncommon but not extraordinary among pair-constrained orderings.
```

--- Sensitivity analysis: reversed bit convention ---

What if bit 0 = top line instead of bottom? All binary values reverse, changing pair types and the difference wave. Key properties tested:

```
Pair structure preserved:  Yes
No-5 property preserved:   Yes
Difference wave identical: Yes
Original entropy: 2.0759, reversed: 2.0759
Original path length: 211, reversed: 211
All key properties are invariant under bit reversal.
```

## Bootstrap confidence intervals

---

Bootstrap confidence intervals

The Monte Carlo results (e.g., '0.18% of random orderings avoid 5-line transitions') are estimates based on sampling. Bootstrap resampling quantifies how precise those estimates are by repeatedly resampling from the results and computing confidence intervals. Narrower intervals = more reliable estimates.

---

```
Base trials: 10,000
Bootstrap resamples: 1000
```

```
No-5-line-transition rate: 0.180%
95% confidence interval: [0.100%, 0.260%]
Interval width: 0.160 percentage points
```

```
Approximately 1 in 556 random orderings
95% CI: 1 in 385 to 1 in 1000
```

Note: These CIs measure the precision of the Monte Carlo estimate, not fundamental uncertainty about the true proportion. They reflect how much the estimate would vary if you re-ran the simulation with 10,000 trials. Increasing --trials narrows these CIs because the estimate becomes more precise.

## Monte Carlo analysis

---

Monte Carlo analysis (10,000 random permutations)

The King Wen sequence has a striking property: no two consecutive hexagrams differ by exactly 5 lines. With 6 lines per hexagram and 7 possible difference values (0-6), is avoiding 5 remarkable or just a coincidence? To find out, we randomly shuffle the 64 hexagrams thousands of times and check how often a random ordering also avoids 5-line transitions. The rarer it is, the less plausible chance becomes as an explanation for the avoidance.

---

```
Permutations with no 5-line transitions: 22/10,000 (0.22%)
Approximately 1 in 454 random orderings share this property.
```

Odds ratio against random: 454:1